

컴퓨터공학과

알고리즘분석 (2026년 3학년 1학기)

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두원공과대학교



CONTENTS

1. Numpy 자료형
2. Pandas 자료형
3. 실습

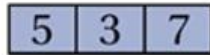


자료 분류

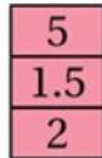
- Scalar
- Vector
- Matrix
- Tensor

(11)

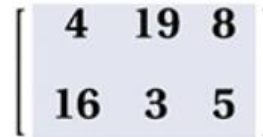
SCALAR



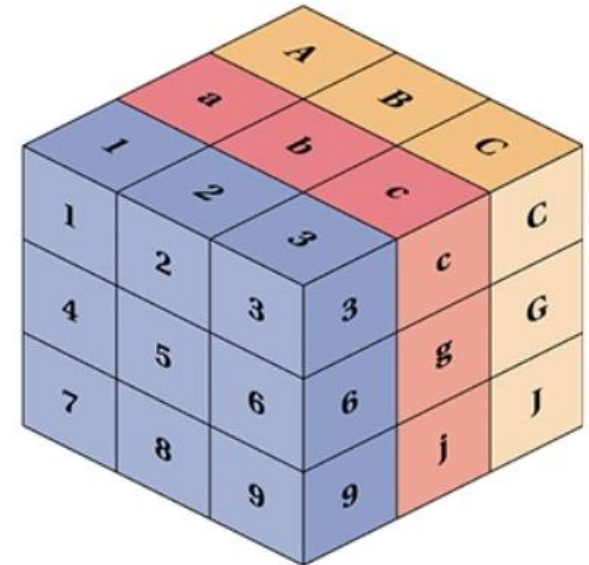
Row Vector
(shape 1x3)



Column Vector
(shape 3x1)



MATRIX



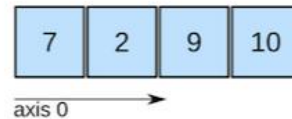
TENSOR



ndarray

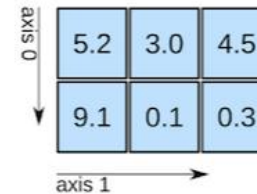
- 형태 (shape)
- 원소의 수 (size)
- 저장된 데이터 타입 (dtype)
- 축 (axis)

1D array



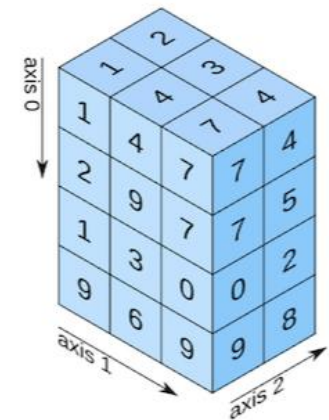
shape: (4,)

2D array



shape: (2, 3)

3D array



shape: (4, 3, 2)



Series vs DataFrame

- Series : 인덱스 index와 값 value로 구성
- DataFrame : 행과 열로 이루어진 2차원 자료구조

Series			Series			DataFrame		
	apples			oranges			apples	oranges
0	3	+	0	0	=	0	3	0
1	2		1	3		1	2	3
2	0		2	7		2	0	7
3	1		3	2		3	1	2